

K8s AutoGuard

Local Kubernetes DevSecOps platform for prevention, detection, guarded remediation, and observable evidence.

Project focus	Kubernetes security automation and operational evidence
Delivery model	Protected GitHub pull requests and reproducible local infrastructure
Author	Chaabi Nour El Houda
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Evidence status
Repository, policy, model, and CI evidence is verified. Final runtime observability screenshots and video recording remain intentionally pending until Docker Desktop WSL integration is restored.

Executive Summary

K8s AutoGuard joins identity-aware network policy, admission control, runtime detection, anomaly classification, and narrowly scoped response. The project is designed for learning and portfolio review: every control is versioned, reproducible, and paired with a validation command.

Architecture and Safety

GitHub PR
Kyverno + Trivy gates
KIND + Cilium + Hubble
Falco runtime event
ML classification
Guarded remediation
Prometheus + Loki + Grafana

The response path defaults to dry run. A Cilium isolation policy can be created only for an anomalous, high-risk, high-severity event in the permitted demo namespace. Kubernetes RBAC permits only create, get, and list access to CiliumNetworkPolicies in that namespace.

Implemented Controls

Layer	Implementation	Evidence
Infrastructure	Two-node KIND, Terraform, Ansible	Cilium-ready cluster workflow
Network	Cilium + Hubble policy demo	77 applicable tests, trusted/denied flow
Runtime	Falco modern eBPF	Controlled event and shell detection
Prevention	Kyverno + Trivy	Secure allowed, insecure denied, CI gates
Detection	Isolation Forest dataset + API	520 labelled events, recall 1.00, FPR 0.08
Response	Guarded remediation	Scoped Cilium policy construction tests
Observability	Prometheus, Loki, Grafana	Metrics tests, Helm rendering, dashboard config

Verified Results

- Cilium connectivity validation completed 77 applicable tests and 320 actions.
- Falco detected a controlled file operation and an interactive container shell.
- Kyverno admitted a Restricted-profile Pod fixture and denied an insecure fixture.
- Trivy security gates and Python quality tests passed on protected pull requests.
- The synthetic development benchmark measured recall 1.00, false-positive rate 0.08, and P95 in-process classification latency 38.69 ms.
- Prometheus metric tests, Grafana dashboard JSON, Kubernetes resource YAML, and pinned Helm templates validated locally.

Limitations

The benchmark uses deterministic synthetic security scenarios and is not a production MTTD or MTTR claim. Loki uses disposable local storage and a test schema. Promtail is included for the project brief but is deprecated; Grafana Alloy is the production-oriented successor.

Final Runtime Acceptance

Step	Command	Evidence to capture
Deploy APIs	<code>./scripts/deploy-autoguard-platform.sh</code>	Healthy ML and remediation Pods
Install observability	<code>./scripts/install-observability.sh</code>	Healthy monitoring and Loki Pods
Run demo	<code>./scripts/run-final-project-demo.sh --interactive</code>	Metrics, alerts, Grafana, Loki logs
Record video	Follow docs/demo/final-project-demo.md	6-8 minute current-session recording

Repository: github.com/chaabi8noor/k8s-autoguard